

Motion App

Motion Runtime Environment for ctrlX CORE
Version 4.6.0

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DC-AE/PAD-SYS (MiSc/MePe)

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1 About this documentation

Editions of this documentation

Edition	Release date	Note
01	2025-11	Motion App version 4.4.0
02	2026-03	Motion App version 4.6.0

Further information

- See [↗ Motion App, Application Manual](#)
- See [↗ Python Runtime App, Application Manual](#)

2 Important information

2.1 Information about our products

For the latest information on our products, go to [↗ https://www.boschrexroth.com](https://www.boschrexroth.com).

For current security advisories, please refer to [↗ https://psirt.bosch.com/security-advisories/](https://psirt.bosch.com/security-advisories/)

3 Version 4.6.0

3.1 New functions

3.1.1 Feature 558466: The checksum function is used to shorten the starting time in the axis profile

Description

The checksum function is used to shorten the starting time in the axis profile. If the drive parameters are not changed, they will not be read again upon the next start. This shortens the starting time.

The checksum is activated if "use-checksum" is set to "TRUE".

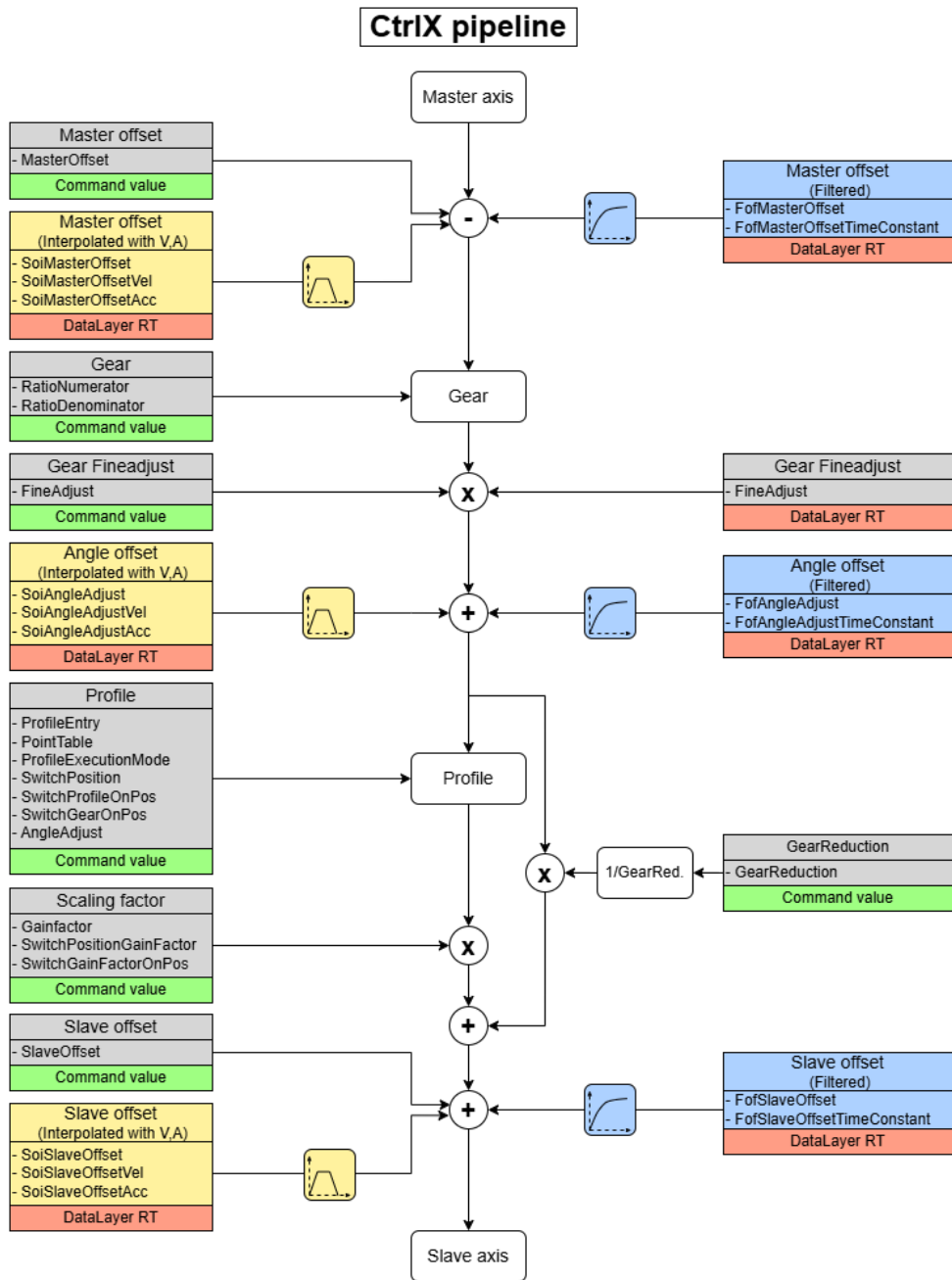
Data Layer path	Information
axisprofile/profiles/<name>/state/device-app-data/use-checksum	Defines/indicates whether the checksum for the profile is activated or deactivated. Data type: Boolean
axisprofile/profiles/system/use-checksum	Defines/indicates whether the checksum for all profiles is activated or deactivated. Data type: Boolean
axisprofile/profiles/<name>/state/device-app-data/checksum-value	Showing checksum value of the profile. Data type: uint32
axisprofile/profiles/<name>/state/device-app-data/checksum-list	Showing list of checksum IDs of the profile. Data type: String array

ctrlX DRIVE SoE/CoE with a firmware version higher than 5.2 and IndraDrive SoE support this function.

3.1.2 Feature 571837: Additional linear offset path to the cam profile

Description

An additional linear path offset was added to the "CamIn" command. It allows to set a gear reduction value. It acts like a gearbox that uses the axis position in front of the cam profile. The result is added up after the cam profile computation.



3.1.3 Feature 656152: Jogging a kinematics in the ACS

Description

The jog commands for kinematics were extended to include jogging in the ACS (axis coordinate system). This allows kinematics to be moved out of a singularity without resolving it.

3.1.4 Feature 815154: Support for "FineAdjust"/"AngleAdjust" register control

Description

"MB_RegisterControllerType01" is extended by new "ControlTypes" (MB_CORRECTION_TYPE):

0 CORR_POSITION: Additive position of the slave axis.

- 1 CORR_ANGLE_OFFSET: Angular offset at profile start.
- 2 CORR_FINE_ADJUST: Fine adjustment of the transmission ratio.
- 3 CORR_ADDITIVE_MASTER_POS: Additive position of the master axis.
- 4 CORR_ADDITIVE_VEL: Additive velocity command value.

3.1.5 Feature 847952: Extension of axis name, axis profile name and kinematic name to 63 characters

Description

The limitation of the number of characters for axis name, axis profile name and kinematic name was increased to 63 characters.

3.1.6 Feature 967203: Wrap-around kinematics

Description

The 2-axis wrap-around kinematics consists of the two stationary rotary axes Rot1 and Rot2, a toothed belt guided around the kinematics via deflection rollers, and a movable vertical center beam. The two rotary axes drive the toothed belt and this moves the movable vertical center beam in the xz-plane.

3.1.7 Feature 1002529: Actual torque additionally in "%"

Description

The actual torque is also output in "%". The maximum torque of the drive is the base.

3.1.8 Feature 1040241: Extension of real-time axis information in the "CXA_PLCOpen" and the "CXA_MotionInterface"

Description

The status parameters drive diagnostics S-0-0390 (CyclicStatusData), actual torque (ActualTRQ) and torque in percent (ActualTRQPercent) were added to the real-time axis diagnostics. These are visible in "arAxisStatus[]". Furthermore, the function blocks "MC_ReadAxisInfo_RT" of "CXA_PLCOpen" were extended accordingly.

3.1.9 Feature 1054135: FAK command from SMC: The step constant of an encoder axis can always be changed

Description

If the encoder parameter "fine-adjust" is set to "TRUE", the fine adjustment value of an encoder axis can be edited at any time:

/motion/axs/<name>/cfg/functions/encoder/fine-adjust

The fine adjustment reads a real-time input value:

- RT data can be written in the Data Layer under the following path:
Motion/axs/[axis_name]/cmd-input/double/data/in.
- Index 10, fine adjustment value (as a unitless percentage value) with the default value 0.0.

3.1.10 Feature 1072777: Increasing the number of cams to 250

Description

Number of usable cams was increased from 99 to 250.

3.1.11 Feature 1088661: Add "MB_TraverseControlType01" to the "cxa_winder" library

Description

"MB_TraverseControlType01" was added to the PLC library "cxa_winder". The function block controls a translatory traversing axis to move different materials together with a winding axis (rotary axis).

During winding, the axis velocity is linked to the velocity of the winding machine and ensures that the material is always traversed tightly and in parallel. The following functions are available for unwinding. They can be selected depending on the requirements (e.g. existing hardware/load cells):

- "MIDDLE_FIXED": The traversing axis is always fixed in the center position.
- "WINDING_STEP": The traversing axis always moves around the material thickness (as when winding).
- "WIRE_TENSION_TRIM": The traversing axis is scaled according to the difference between the actual tension value of the traverser and the actual tension value outside the traverser and is traversed with a PID controller (two load cells are required).
- "WINDING_STEP_TENSION_TRIM" - Combination of the methods 2 and 3. It is traversed according to the material thickness and the tension difference.

3.1.12 Feature 1095638: Configurable singularity handling

Description

Singularity handling can now be deliberately disabled. If this setting is selected, the kinematics stops if a singularity is detected on the path and an error message is sent when preparing the motion.

3.1.13 Feature 1095701: Warning note near singularities

Description

If a singularity is detected on the programmed path, a warning can now be displayed containing information about the position, orientation and axis with the strongest motion at the singularity.

3.1.14 Feature 1097156: Half-jump motion

Description

The half-jump combines an upward or downward motion with a horizontal or synchronization or desynchronization motion. It is often used together with the belt-synchronous motion.

3.1.15 Feature 1097220: Blending belt-synchronized motions

Description

This function allows to blend belt-synchronized motions. This enables faster and smoother motions when synchronizing between two belts or a late change of the target position.

3.1.16 Feature 1102854: Validation command for large point tables

Description

For large point tables with more than 1024 entries, it takes too long to check the continuity of the progression. Thus, a new validation command (axsCamInvalidate) was introduced that performs the check in advance.

3.2 Resolved defects and modifications

3.2.1 Bug 1087691: If the Motion app is in configuration mode and the AxisInterface starts, there is no feedback for the user

Description

If the Motion app is in configuration mode and the AxisInterface starts (initialization is running), "TE_AxisInterfaceMainProg.InitDone" cannot be reached without an error or message.

Troubleshooting

"TE_AxisInterfaceMainProg.InitDone" remains set to "FALSE", and the additional information "Wait for Motion switching at runtime" was added to "Global_AxisInterface.AdditionInfo".

3.2.2 Bug 1088822: Add the new global variable "AddtionInfo" in "Global_AxisInterface" of the library "CXA_MotionInterfaceUser"

Description

If "GlobalAxisDefines" is used for axis configuration and "GlobalAxisDefines" differs from the Motion configuration in the Motion Runtime app, the initialization fails and the general error code 090F0053/0C570101 is output. However, troubleshooting is difficult due to the general error message.

Troubleshooting

Add the new global variable "AddtionInfo" in "Global_AxisInterface" of the "CXA_MotionInterfaceUser" library. If there is an error, the message displays user-defined information, e.g. which axis is affected. If an error occurs, the message contains further user-defined information, e.g. which axis definition is faulty.

If the name of the axis is incorrect or the axis does not exist, the additional message "AxIf Init: could not read cfg from" is displayed, which is applied from "CXA_MotionInterface.MB_AxisInit.DiagStatus.Message".

3.2.3 Bug 1095474: Diagnostics "Position jump in an axis" detected at the start or end of a command

Description

The diagnostic message "Position jump in axis..." can be issued. This mainly occurs at the start or end of a command if the relation between the path value and the axis value is highly non-linear. At these points, computations have so far been performed with insufficient accuracy.

Troubleshooting

Special "treatments" now take place at the start and end of a command to ensure a more accurate computation.

3.2.4 Bug 1097524: Error "Start plane is violated in negative direction" occurs during "KinBeltSync" synchronization

Description

The machine is on the "Start plane" and tries to synchronize to a belt with "KinBeltSync". The error message "Start plane is violated in negative direction" can be issued. This is due to the fact that in the first few cycles of synchronization, some compensations can briefly push the motion slightly in front of the "start plane".

Troubleshooting

The check was too strict and is now only performed in the "KinBeltSync" state "Synchronized".

3.2.5 Bug 1111967: Dynamic axis limits are violated when the "KinBeltSync" command option is active and when axis transformation is active**Description**

When synchronizing to a belt with "KinBeltSync" and active non-linear axis transformation, for example with a 6-axis robot, the dynamic axis limits may be violated depending on the current position in the workspace.

Troubleshooting

The dynamics used for synchronization and desynchronization are now taken from a point where the axis transformation is taken into account. As a result, both synchronization and desynchronization, as well as the trajectory dynamics of the kinematics, are further reduced.

3.2.6 Bug 1112046: Exception when using synchronized Motion commands**Description**

If the user executed a synchronized Motion command, e.g. a FlexProfile, and changed the real-time inputs for the synchronized commands (e.g. master offset) from a PLC task running simultaneously, an access violation could result.

3.2.7 Bug 1116140: Synchronization could not be performed in case of a belt standstill**Description**

Error in "BeltSync" if the belt was at a standstill and the synchronization position was behind the start level.

3.2.8 Bug 1131683: If "KinCyclicSetPointPCSCmd" is active, "KinCoordTransform" always returns 0 for position and orientation when moving from or to the WCS**Description**

Position and orientation capabilities were not set when using the "KinCyclicSetPointPCSCmd" command.

3.2.9 Bug 1134071: The states "Data.VelMode.InVel" and "Data.DeviceStatus.Invelocity" are inconsistent**Description**

Sometimes the states "Data.VelMode.InVel" and "Data.DeviceStatus.Invelocity" were inconsistent.

3.2.10 Bug 1135255: The power-on of a drive could cause operation mode switching error**Description**

Non-cyclic switching of the operation mode when switching on the power of a drive caused an error.

3.2.11 Bug 1138167: General kinematic command error with motion command after reset

Description

After a kinematic reset, a general kinematic command error could occur if a motion command was executed immediately afterwards.

3.2.12 Bug 1138758: Incorrect behavior with deactivated axis/kinematics

Description

If an axis/kinematic is disabled via "motion/axis (kin)/<name>/cfg/properties/activated = FALSE" and the user attempts to switch on this axis/kinematics, the wrong error code for the axes 090F2002/0C570112 and for the kinematics 0A0F0107/0C230155 are displayed on the "MotionInterface" user interface.

Troubleshooting

If the axis/kinematics is disabled via "motion/axis (kin)/<name>/cfg/properties/activated = FALSE", all commands for the axis/kinematics are disabled and no error code is displayed.

3.2.13 Bug 1139189: The error reaction of an axis can trigger at higher limit values than the configured ones

Description

If an axis was in motion and it was detected that the axis limits were exceeded, it was possible that the axis triggered at higher dynamic limit values than those configured for this case.

3.2.14 Bug 1141031: Inserting the "AxisInterface" template does not work correctly

Description

If the "AxisInterface" template was inserted, the wrong program was inserted in the "MainTaskMotif" task. The program "AxisKinIfPlcProg" was inserted there instead of "AxisIfPlcProg", so that the project could not be compiled without errors.

3.2.15 Bug 1142883: Blending together with continuous motion

Description

Command option "Blending" - command has final velocity, but cannot execute the next command due to active "blending" with the predecessor command.

Troubleshooting

If an error occurs, an error is generated and an error message is displayed.

3.2.16 Bug 1143245: Filters of axes are always active

Description

The filters for the actual values of an axis are always active. This is to be changed to reduce the computing load of the cyclic calculations.

Troubleshooting

The following logic was implemented for the filters: The filter type "NONE" deactivates the filters of an axis. Other filter types activate the axis filter.

The filter can only be activated/deactivated in the "Configuration" state of the Motion ("NONE" <-> "ANY").

It is always possible to change between the active filter types (all except "None").

3.2.17 Bug 1145458: FPU arithmetic error: "Zero division" in the "GearInVel" Motion function

Description

Changing the master offset using RT inputs could lead to the FPU error "zero division" in the "GearInVel"/"GearInVelEx" functions.

3.2.18 Bug 1153417: "MoveAbs" fails with an error in the job list

Description

Internal error that prevented the command from being executed.

3.2.19 Bug 1156941: Parameter of "SwitchOnPath" used is not displayed in the trace

Description

After activating the "SwitchOnPath" command option with parameters for the target kinematics, the "Trace" messages and the Data Layer nodes for pre-selected (motion/kin/RB002/state/diag/cmd-opt-preselected) and active command options (motion/kin/RB002/state/diag/cmd-opt-active) only displayed the name of the "JobOptSwitchOnPath" command option without parameter details.

3.2.20 Bug 1157207: Improvement of the diagnostic information on the absolute position of the command

Description

Previously, the diagnostics did not include the required diagnostic code and command name to localize the error.

3.2.21 Bug 1158319: Position jumps during "BeltSync" resynchronization with different belt directions.

Description

A position jump in one of the axes involved occurs when the motion is resynchronized from one belt to another with a different belt direction using "BeltSync". The system offsets parts of the compensation in the wrong direction.

Troubleshooting

The system now handles certain compensations separately and applies them in the correct directions.

3.2.22 Bug 1161622: The Interrupt mode of "MotionInterface" cannot stop the motion executed via the Motion user interface

Description

After the ctrlX CORE changed to "Running" or a PLC reset command was executed, the interrupt command executed by "CXA_MotionInterface" reported the error 0A0F0107/0C230155 when the axis was in a motion triggered by the Motion UI and could not stop the motion.

3.2.23 Bug 1161936: "Python-SyncMotion" commands do not transfer the "fine_adjustment" parameter correctly

Description

"Python-SyncMotion" commands did not transfer the "fine_adjust" and "fine_adjust_dl_idx" parameters correctly to the Motion kernel. 0 was always effective.

- axs_cmd_gear_in_pos
- axs_cmd_gear_in_pos_ex
- axs_cmd_gear_in_vel
- axs_cmd_gear_in_vel_ex
- axs_cmd_flexprofile
- axs_cmd_flexprofile_ex
- axs_cmd_cam_in
- axs_cmd_cam_in_ex

3.2.24 Bug 1162010: Position jumps during "BeltSync" resynchronization with different belt directions

Description

A position jump in one of the axes involved occurs when the motion is resynchronized from one belt to another with a different belt direction using "BeltSync". The system offsets parts of the compensation in the wrong direction.

Troubleshooting

The system now handles certain compensations separately and applies them in the correct directions.

3.2.25 Bug 1163663: Commanding a KinMove3DCircle with a 2D transformation

Description

Commanding a 3D circle with a 2D transformation triggers the error that a parameter is missing without specifying the missing parameter.

Troubleshooting

Commanding a 3D circle with a 2D transformation is now possible and the error message indicates which parameter is missing.

3.2.26 Bug 1165815: Target position violates axis position limits when "BeltSync" is active

Description

If "BeltSync" is active, the error message "Target position violates axis position limits" can occur when preparing a motion command. The cause is that the motion is prepared at a different point in the axis space than where it is actually executed later. At the time of preparation, the exact motion in the axis space is not yet known when "BeltSync" is active.

Troubleshooting

The check of the axis position limits at the time of preparing a command is now deactivated if the "BeltSync" command option is active. To ensure compliance with workspaces, checks should instead be configured during the motion processing.

3.2.27 Bug 1166004: Error code is not detected if "MC_Power off" is executed during an axis motion**Description**

The "MC_Power" deactivation function of CXA_PLCopen is only valid if the PLCopen state is in "STANDSTILL" or "STANDSTILLPENDING". In other states, the axis retains its status and no error message is displayed. This can cause problems.

Troubleshooting

By activating/deactivating "MC_Power", the "PLCopen" state of the axis changes from "DISABLED" to "STANDSTILL" or from "STANDSTILL" to "DISABLED". Deactivating "MC_Power" from another "PLCopen" state, e.g. "Continuous_Motion", causes an error state with the error code 091F2002/0C570100.

3.2.28 Bug 1167996: Missing phase end condition when computing the velocity profile**Description**

Under certain circumstances the 4th phase end condition might not have been computed. This could cause an error when computing the other phase end conditions.

3.2.29 Bug 1170569: Error when preparing the IVAJ interpolator - no valid solution found**Description**

In exceptional cases, the interpolator did not find a suitable solution when computing the velocity profile.

3.2.30 Bug 1171630: The subroutine cannot access the global variable defined in the main script**Description**

If a global variable was defined in the main script, it could not be accessed in the subroutine.

3.2.31 Bug 1180519: Error in the kinematic interpolator (IVAJ) while preparing a "KinJog" command**Description**

An error occurred in the kinematic interpolator (IVAJ) while a "KinJog" command was being prepared. This caused the error message described with the following detailed diagnostics: Error in kinematic interpolator IVAJ. prepare() returns with result: -267255807.

Troubleshooting

The internal logic for the command preparation in the IVAJ interpolator was revised and is now more robust to resolve the problem.

3.2.32 Bug 1180767: Exceptions occur in the PLC logger when the PLC application logs on and off

Description

Exceptions occur in the PLC logger when the PLC application logs on and off. The memory was overwritten, as the number of iterations of the FOR loop exceeded the number of elements in the "afixedValue" array.

Troubleshooting

Correct the number of iterations of the FOR loop to prevent the memory from being overwritten.

4 Version 4.4.0

4.1 New functions

4.1.1 Feature 265408: Adjusting the drive and control configuration

Description

This interface function enables the configuration of an SOE drive to be compared with the configuration in the ctrlX MOTION in the control system. Both directions are supported. Axis limit values are automatically configured slightly higher in the control than in the drive.

4.1.2 Feature 523852: Command dwell time and CNC command G4 added

Description

The dwell time for motion commands is implemented in the Motion app and the CNC command G4 is added to the G-code app.

Example in CNC:

```
N10 G04 F10 //Waiting for 10 second
```

4.1.3 Feature 553369: GearInPos: Dynamic synchronization to a defined master axis position

Description

This function enables the synchronous state of the slave axis to be reached at a defined master axis position. The synchronization process of the slave axis is completed when the defined master axis position is reached.

4.1.4 Feature 613504: Extended number of SoMO signals per kinematics

Description

In addition to the 100 global signals, 1000 local signals are added for each kinematics. These signals are stored in kinematics-specific Data Layer nodes.

DL path for global signals: "motion/state/functions/somo/rt-data/".

DL path for kinematic signals: "motion/kin/*/functions/somo/rt-data/data".

Motion commands:

- "setLocalSignalCmd": sets a local kinematic signal to "true"
- "resetLocalSignalCmd": sets a local kinematic signal to "false"
- "waitForLocalSignalCmd": waits until a local kinematic signal becomes "true".

- "waitForLocalSignalResetCmd": waits until a local kinematic signal becomes "false"
- "setLocalSignalCmdOpt": sets a signal once, triggered by the following motion command

4.1.5 Feature 614717: Introducing a WCS (World Coordinate System)

Description

This functionality introduces the affine transformation between MCS (machine coordinate system) and WCS (world coordinate system). It facilitates global placement of kinematics in space. Each kinematic system has its position and alignment in the WCS, but can still work locally in the MCS.

4.1.6 Feature 651031: Extending the points tables to 100,000 entries

Description

The maximum size of point tables has been increased from 1024 to 100,000 entries. The FlexProfile uses point tables as CAM tables to synchronize complex axis motions. Switching between large point tables is possible in real time. Loading and checking the tables takes more time and is therefore not real-time capable.



The use of table points greater than 1024 (LAW_TYPE 'LINEAR' or 'SPLINE') is only possible with the FlexProfile command (cam table within a segment). Use with the CamIn command is only possible with tables with up to 1024 table points (LAW_TYPE 'DEFAULT').

4.1.7 Feature 682221: Cyclic commanding of target values of kinematics

Description

This function disabled the axis-synchronous control of a kinematics. For example, a virtual axis can control the velocity of a robot synchronously with other motion functions of the machine.

4.1.8 Feature 741724: Configurable axis error response

Description

The standard error response is not flexible enough to achieve the best possible response for all error cases and all machine types. It is often necessary to adjust the error response in special cases (e.g. for a SyncMotion axis if a drive error F2 occurs).

The user can now define an error response for each error class. In addition, an error response can affect a group of several axes.

4.1.9 Feature 759059: Kinematic motion for pick-&-place applications

Description

The command generates the following motions:

- Upward motion (based on the specified starting height parameter)
- Motion on a polynomial path into a horizontal motion (based on the specified maximum height parameter)
- Horizontal motion (based on the specified target position)
- Motion on another polynomial path into a vertical motion downwards (based on the specified final height parameter)
- Completion with a vertical downward motion (until the target position is reached)

4.1.10 Feature 770497: Continuous monitoring of the axis reference**Description**

Previously, the axis reference was only checked when the power was switched on. The check is now carried out in all axes states. This affects Motion commands that require a referenced axis.

4.1.11 Feature 842957: PathSmooth option for transition in PLC kinematic function blocks**Description**

The "TransitionMode" input of the MC_MoveLinearAbsolute, MC_MoveLinearRelative and MB_MoveCircle2D function modules has been expanded to include the new "TMPathsmooth" option. This option inserts a soft, polynomial-based transition between two commands.

4.1.12 Feature 847043: Orientation with commanded and interpolated axis values for orientation axes**Description**

Many applications require direct commanding and interpolation of the axis orientation. This is particularly the case for applications with 1D orientation, but often also with 2D orientation (5-axis machines).

A major benefit of this method is that the axis motion is more predictable for the user (direction, position and limit values). Furthermore, there are no problems with singularities.

4.1.13 Feature 880248: Extending the axis profile with the homing status**Description**

After configuring the signal status word (S-0-0144) in ctrlX I/O Engineering and selecting "Homing status: feedback value status" (S-0-0403.0) in ctrlX DRIVE Engineering, the drive parameter home status can be updated as cyclic axis data in the Data Layer node "motion/axs/[axis name]/state/values/homed".

4.1.14 Feature 882521: Working area monitoring of the TCP**Description**

This functionality permanently enables the monitoring of an individual working area for a kinematic system. It represents an extension of the working areas and safety zones.

4.1.15 Feature 905303: NC commands RAX, WAX and GAX are added in the G-code app**Description**

The NC commands RAX, WAX and GAX are added to the G-Code app:

- RAX: Remove the axis from the kinematics
- WAX: Wait until axis is in idle state, then add to kinematics
- GAX: Add the axis to kinematics if it is in idle state

4.1.16 Feature 914813: Implementing the axsEmergencyStop motion command

Description

This command sets the error level to SEVERE and indirectly enables the error response process (if the axis is connected to a drive, the drive stop function is enabled at the same time, otherwise only the interpolation delay of the control).

4.1.17 Feature 956295: Switchable monitoring of whether an axis follows the position specification

Description

For some applications, it is necessary to switch off the monitoring of whether an axis is following the command values. One example is checking the brakes in the drive.

The axsSetDriveFollowsMonitoringCmd command can be used to switch monitoring off and on again.

4.1.18 Feature 973427: Belt Sync: Relative desynchronization

Description

This function introduces relative desynchronization for belt synchronization. In contrast to absolute desynchronization, it takes the belt motion into account. The target position is therefore relative to the belt position.

4.1.19 Feature 974228: Improving the MExA SDK

Description

Build and artifacts have been corrected and standardized. The documentation has been extended.

4.1.20 Feature 983590: Soft stop for Motion commands

Description

To avoid errors with Motion objects while the script is to be canceled, the script manager supports a stop command. This command sends the SIGTERM signal to the Runner process. In Python, the user has the option of installing a signal handler for certain clean-up functions. In this clean-up process, for example, Motion commands can be canceled and the Motion objects can be separated. Subsequently, the signal handler can then end the runner without a Motion error.

Example:

```
# stop command handler
def sigint_handler(signum, frame):
    if signum == signal.SIGTERM:
        print("SIGTERM received, exiting via sys.exit().")
    else:
        print(f"unknown signal {signum} received.")
        sys.exit()

# cleanup motion
motion.axs_cmd_abort("Axis_1")
motion.axs_cmd_abort("Axis_2")
motion.detach_obj("Axis_1", "Axis_2")

# exit script normal
sys.exit(0)
```

4.1.21 Feature 992141: Extending CXA_MotionPrinting with the MB_TensionControlLoadCellMatParType01 function block

Description

Depending on the measured value of a load cell, the MB_TensionControlLoadCellMatParType01 function block regulates the web tension to the command value. Two function modules are included in the PLC library CXA_MotionPrinting:

- Public function block for users: MB_TensionControlLoadCellMatParType01 provides a voltage controller with internal controller setting
- Internal function block: MB_TensionCalcControlParamType01 calculates the PID parameters of MB_TensionControlLoadCellType01 based on the web material thickness

4.1.22 Feature 996126: Disabling axes for modular machines

Description

An axis or kinematics not available in the current machine configuration can be disabled. A disabled axis or kinematics is still in the project. All configuration options can still be addressed. The machine runtime is not negatively affected.

The new axis and kinematics configuration becomes active when switching from SETUP to OPERATION.

4.1.23 Feature 1007148: Script manager: Command to display a message in the logbook

Description

The user can create their own messages, warnings and errors using Python or G-code commands. This information is displayed in the diagnostic log.

Syntax in G-code:

```
SET_MSG("My gcode message")
SET_WARN("My gcode message")
SET_ERR_F0("My gcode F0 ERROR")
```

Syntax in Python:

```

userlog.print_msg("my message")
userlog.print_warn("my warning")
userlog.print_err_f0("my f0 error")

```

4.1.24 Feature 1037798: Window for "InSynchronization"**Description**

In addition to the status message "InSynchronization" (the slave axis command value has reached the master axis command value), an "InSyncWindow" status message is also implemented (both in the cyclicStatusData structure of the respective axis, which can be enabled for an axis in the "real-time status data").

A sync window position and a sync window velocity are available for configuring this new "InSyncWindow" status message. The Sync-Window-Position is valid for a Sync-Position command (e.g. GearInPos) and the Sync-Window-Velocity is valid for a Sync-Velocity command (e.g. GearInVel).

The "InSyncWindow" status message is set if the difference between the master command value and slave command value is < of the configured window.

4.1.25 Feature 1042796: Tension Control - MB_TensionControlDecouplingNetworkType01**Description**

The CXA_MotionPrinting library has been expanded to include the following function modules:

- MB_TensionloadcellType02
- MB_TensionControlDecouplingNetworkType01

Both function blocks provide a solution for setting up/decoupling a voltage control network.

4.1.26 Feature 1046076: ML_KinBeltSyncP has been integrated into the PLCopen library**Description**

The MB_KinBeltSyncP function block enables and disable the command option for the path-synchronous motion.

4.1.27 Feature 1051941: Extended kinematic interface is expanded to include PathSmooth mode**Description**

The "PathSmooth" mode has been added to the kinematic interface. The new property "ActivePathSmooth" has been added to "MB_KINEMATICS_CMD_OPTION" to switch the "PathSmooth" function on and off. When this function is enabled, it inserts blending or rounding using a polynomial path between all consecutive Motion commands.

4.1.28 Feature 1062733: G-code is extended to local variables

Description

In addition to global variables, G-code also supports local variables with dynamic data types. Local variables are defined with the keyword "LOCAL" and can be specified with/without an initial value. The scope of validity of local variables is limited to a file or a block, as shown in the example program.

Example:

```
startPos_X = 10                // startPos_X: [10,
                                global]
LOCAL startPos_Y = 20          // startPos_Y: [20, local
                                in sample program]
FOR INDEX = 1 TO 3 DO
  LOCAL posX = startPos_X + INDEX // posX: [local inner For
                                      Statement]
  LOCAL posY = startPos_Y + INDEX // posY: [local inner For
                                      Statement]
  POS = posX + posY             // POS: [global]
ENDFOR
```

4.2 Resolved defects and modifications

4.2.1 Bug 965196: "InPosition" is only set with absolute positioning

Description

There is an "InPosition" bit in the cyclic axis data. This is not set correctly for relative or additive motion commands. This was caused by the fact that the commanded position was used internally to generate the bit.

Bugfixing

The absolute target position is used internally to generate the "InPosition" bit.

4.2.2 Bug 1001692: Some diagnostics for belt synchronization are sent without SourceInfo

Description

Some diagnostics for belt synchronization were sent without SourceInfo.

4.2.3 Bug 1003568: The power connection of a drive could result in an operating mode switching error

Description

Non-cyclical switching of the operating mode when switching on the power of a drive resulted in an error.

4.2.4 Bug 1008015: SyncMotion slave axis with jerk limiter travels too far

Description

A SyncMotion slave axis could travel too far if a low jerk limit was configured for it and the master axis changed its velocity with a very steep ramp and the slave axis was in the SYNCHRONIZED state at the time.

4.2.5 Bug 1016439: Error in limit switch monitoring for SyncMotion axes

Description

If a SyncMotion slave axis was moved with very high dynamics (as the master axis was commanded accordingly) and the motion was very close to the axis limit switch, an error could occur in the monitoring of the axis limit switches. An error could be detected even though the motion was correct. As a result, an error reaction was started. A similar problem could occur with Gantry axes.

4.2.6 Bug 1017474: Signature conflict when using 3.4.0.x CXA_Motion project with a new CXA_Motion 3.6.0.12 library

Description

When using a PLC project that was created with a CXA_Motion 3.4.0.x library, a signature conflict occurred if the new PLC library CXA_Motion 3.6.0.12 was used. The problem concerned the following functions: ML_iKinPCSP, ML_iKinToolLengthsP and ML_iKinMCSP.

4.2.7 Bug 1019168: Error in limit switch monitoring in an error response

Description

If an axis is moved with very high dynamics (as the master axis is commanded accordingly) and the motion is very close to the axis limit switch and an error occurs during the braking ramp, an error may occur in the monitoring of the axis limit switches. An error is detected even though the motion is correct. As a result, an error response is started which can exceed the limit switches.

Bugfixing

Either travel with a (slightly) reduced dynamics when you are close to the limit switch, or increase the configured dynamic limit of the axes.

4.2.8 Bug 1025243: CamIn: Incorrect Shannon error

Description

When controlling a CamIn with a gainFactor < 360, a Shannon theorem error could occur even though the velocity for the cycle time was not too high.

4.2.9 Bug 1030818: CamIn: Incorrect position after switching synchronization off and on again

Description

When synchronizing to a master with CamIn, then canceled the synchronization and synchronized again with the same CamIn, this could result in an incorrect position.

4.2.10 Bug 1037835: FlexProfile change: Incorrect synchronization behaviour

Description

Two problems occurred when switching from one FlexProfile to another FlexProfile:

1. ➤ Even if the FlexProfile was commanded with "ML_DynSyncDirection.DIRECT", the slave axis did not remain in the SYNCHRONIZED state if the axis limit values of the slave axis were exceeded.

2. ➔ Even if the FlexProfile was commanded with a "SwitchPosition", synchronization started immediately and not only when the "SwitchPosition" was reached.

4.2.11 Bug 1045397: AH/AF is enabled if the PLCOpen status is "DISABLE"

Description

AH/AF was green when the ExternalFB mode is enabled and the PLCOpen status was at "DISABLE", but should be gray to indicate that the device was off.

4.2.12 Bug 1047427: Error message when saving the Motion AppData

Description

When saving the Motion AppData, a timeout error could occur if saving took more than 10 seconds. This can occur if many axes are created and virtual control is used. The data is saved correctly despite the message.

4.2.13 Bug 1052417: AxisInterface: Structure element "IpoAddValues" cannot be used

Description

All elements of the "arAxisStatus_gb[.Data.IpoAddValues" folder did not work with the AxisInterface.

4.2.14 Bug 1054295: Axis coupling for 6-axis robots with spherical hand

Description

There are mechanical designs of the 6-axis robot with spherical hand, in which rotations of one rotary axis result in a coupled rotation of one or more other rotary axes. Coupling factor pairs exist in order to be able to compensate for this behavior by the respective axis affected and thus enable isolated rotations of specific axes.

Bugfixing

Coupling factor pairs implemented for axes 4, 5 and 6:

- cRot5Rot4Ratio1 with cRot5Rot4Ratio2, which describe a coupling from Rot4 to Rot5
- cRot6Rot4Ratio1 with cRot6Rot4Ratio2, which describe a coupling from Rot4 to Rot6
- cRot6Rot5Ratio1 with cRot6Rot5Ratio2, which describe a coupling from Rot5 to Rot6

4.2.15 Bug 1066480: Start position not accessible after switching on the belt synchronization

Description

The start position could not be reached after switching on the belt synchronization due to an internal calculation strategy.

4.2.16 Bug 1066623: Position gap during the KinBeltSync desynchronization command

Description

The position gap diagnostics is sent shortly after the change to the KinBeltSync synchronization command. This error only occurs if an offset in the y- or z-coordinates is configured for the KinBeltSync PCS or if another PCS is activated directly after KinBeltSync is switched off.

Bugfixing

The direction in which the compensations are calculated for the KinBeltSync desynchronization command is now calculated correctly.

Restriction:

Furthermore, no PCS may be activated directly after switching of KinBeltSync, but only after the command of the desynchronization command.

4.2.17 Bug 1067514: The "Done" output of the MC_AddAxisToGroup function block is not set to TRUE

Description

After the function block has been executed, the "Done" output cannot be set to "TRUE". At this point, the PLCopen status is in "CoordinatedMotion".

Bugfixing

Once the function block has been executed, the "Done" output can be set to "TRUE". In addition, the MC_AddAxisToGroup and MC_RemoveAxisFromGroup function blocks are moved from the "Axis Group/POUs/Command" folder to the "Single Axis/POUs/Command" folder. This change is compatible.

4.2.18 Bug 1067687: Unwarranted runtime error with KinAbort in connection with Blending or ContMotion

Description

When KinAbort is commanded, the kinematics switches to STOPPING and starts a deceleration ramp. If a new command was enabled during this braking ramp (due to Blending or ContMotion), an (incorrectly issued) runtime error occurred.

4.2.19 Bug 1087732: "CamIn": Shannon error in "SwitchPosition"

Description

If a "CamIn" was commanded with a "SwitchPosition" where a modulo overflow occurred in the slave axis and with SLAVE_ORIGIN_MASTER_*, a Shannon error occurred even if the transition was smooth and was completed successfully.

4.2.20 Bug 1088860: "Flexprofile"/"CamIn": Acceleration peak on command change

Description

If a FlexProfile was active and another FlexProfile was commanded during an acceleration phase of the slave axis, an acceleration peak could occur (even if a "switch position" was commanded).

4.2.21 Bug 1089728: IL_ModbusTCPReadDiscreteInputs: An error occurs with a quantity > 320

Description

IL_ModbusTCPClient generates the error code 16#0C23020E after IL_ModbusTCPReadDiscreteInputs has been enabled with an input quantity of more than 321.

Bugfixing

IL_ModbusTCPClient now supports 2000 signals and the "Quantity" input of IL_ModbusTCPReadDiscreteInputs can be set to 2000.

4.2.22 Bug 1098599: "CamIn": Inaccuracy in "SwitchGainFactorOnPos"

Description

The accuracy of the "SwitchGainFactorOnPos" function is the cycle time sampling. This resulted in an inaccuracy of the calculation of the position delta.

Bugfixing

A new check for the return direction has been added to ensure the correct winding direction and material tension.

4.2.23 Bug 1107138: Error "Drive does not follow command value specification" on telegram failure

Description

The ctrlX MOTION tolerates up to two telegram failures with status data from the drives. However, this was not taken into account when monitoring the status information "Drive does not follow command value specification". This could result in the error 090F2002 0C560104 "Axis cannot (or can no longer) follow..." if there was a telegram failure.

4.2.24 Bug 1110198: Low velocity (standstill mode) in reverse direction results in wrong direction

Description

In the return direction, a low velocity results in a wrong direction, which can reduce the tension on the material.

4.2.25 Bug 1118239: Subroutine path node does not work correctly

Description

The default subroutine path cannot be changed via the Data Layer. When attempting to delete part of the path, the string is always incorrectly truncated at the previous slash instead of deleting only the desired characters.

Bugfixing

After saving, the default subroutine path in the Data Layer or JSON file has to display exactly the string entered by the user, without unexpected truncation or automatic trimming of the path.

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